NearWord

How the game works - in simple words, then in the code

1. The game in one minute

A secret word is hidden. You type any word. The app tells you a **rank**: how close your word is to the secret word in **meaning** (not spelling). Rank 1 is the secret word itself. Smaller number = closer. Colours: green = rank 1 to 300, yellow = 301 to 1,500, red = above 1,500.

Example: secret word is **cat**. 'dog' gets rank about 5 (very close), 'house' about 3,000 (far), 'democracy' about 20,000 (very far).

2. Three boxes: where everything lives

Box	What is inside	Needs internet?
Box 1 - built into the app	A dictionary of 40,000 words. Each word has 100 'meaning numbers' (words with similar meaning have similar numbers). Also the list of words that are allowed to be the secret answer (about 1,850) and the category lists.	No
Box 2 - saved on your phone only	Coins, streak, stats, achievements, unfinished games, nickname, theme, haptics. Your friend's phone has its own separate copy. Nothing here is shared.	No
Box 3 - Firebase (online)	Only the live **Battle** rooms: who is in the room and each player's best rank. Right now it is switched OFF (no database URL set yet).	Yes

3. Where does the secret word come from?

There is **no list of 'today's word' online** and no server that picks words. Every phone **calculates** the word itself, and because the calculation is exactly the same, every phone gets the same word.

Mode	Puzzle id (the 'seed')	What it is made from
Daily	daily-2026-10-13	today's date
Quick play / Challenge	rnd-ABC123	a 6-character code
Category	cat-animals-ABC123	category + code
Battle	battle-K7Q2X	the room code

The id goes through a fixed formula (a hash) that always gives the same number. That number picks one word from the built-in answer list.

```
id  ->  hash(id)  ->  number  ->  answers[number % list_size]  ->  the secret word
```

Same word on different devices: same id + same built-in list + same formula = same word. That is why your friend can type your challenge code and get your word, why everyone gets the same daily word, and why Battle never needs to send the word over the internet.

4. How the rank is made (on the phone)

- The app takes the secret word's 100 meaning numbers.
- It compares them with all 40,000 words (a dot product = a similarity score).
- It sorts all words from most similar to least similar.
- A word's position in that sorted list is its rank. The secret word is always position 1.

This takes a fraction of a second, so rankings work completely offline.

5. What happens when you type a word

- The text is cleaned: spaces removed, lowercase, accents removed, and the singular form is tried too ('berries' becomes 'berry').
- Not in the 40,000-word dictionary: 'Word not in our list.'
- Already guessed: the old row flashes, nothing is added.
- Otherwise the word is added with its rank. The list stays sorted best-first. Rank 1 = you win.
- Your progress is saved on the phone, so you can close the app and continue later.

6. What Firebase is and what it does here

Firebase is a free online 'notice-board'. Phones can write short notes on it and read notes written by other phones. NearWord uses it **only for online Battle rooms**. Nothing else (coins, stats, daily word, rankings maths) goes to Firebase.

When you create a battle room, the phone writes a note like this:

```
rooms / K7Q2X
    host:      you
    status:    waiting   (host presses Start -> playing -> finished)
    duration:  180 seconds
    players:
        you:     best rank 1114, 5 guesses, solved: no
        friend:  best rank  320, 8 guesses, solved: no
```

Step by step:

- **Create:** your phone writes the room note and joins as the host.
- **Join:** your friend types the 5-letter code; their phone adds its own row under 'players'.
- **Start:** the host sets status to 'playing'. Everyone sees a 5-second countdown.
- **Play:** every phone works out the SAME secret word from the room code, then computes its own ranks locally. After each guess it updates only its own row (best rank, guess count).
- **Scoreboard:** every phone reads the whole note about once a second and sorts it with the same rule, so everyone sees the same live ranking: solved first (fastest wins), then nearest rank, then fewest guesses.
- **End:** when time runs out, everybody finishes, or 30 seconds after the first person solves.

Why it is cheap and simple: only tiny notes travel; the secret word is never sent; there is no server code to run. It uses Firebase's web (REST) address, so no Firebase SDK files are needed.

To switch it on: create a Firebase Realtime Database, paste its URL into lib/core/constants/backend_config.dart, and paste database.rules.json into the Rules tab. Until then 'Practice vs bots' works offline with fake players.

7. Daily puzzle with a friend - do you both see names in a rank?

Not yet. Today the Daily puzzle is solo: you and your friend get the **same word** (same date = same seed), but each plays alone and there is no shared daily scoreboard. Only **Battle** shows player names and a live ranking.

What a real daily tournament would need (not built yet):

- When a player finishes, write one small note online: daily / 2026-10-13 / name -> guesses, time, solved.
- A 'Today's leaderboard' screen that reads those notes and shows everyone's name, guesses and time, best first.
- Optional friends-only version using a code, like battle rooms.
- It needs the Firebase URL set up first (same free database).

8. Code map: which file does what

Question	File (inside lib/ unless noted)
Where do the words and meaning numbers come from?	tools/build_embeddings.py (run once on a PC). Produces assets/data/vocab.txt, vectors.bin, secrets.txt, categories.json. These are bundled in the app.
How does the app load them?	core/services/semantic_engine.dart - init() loads, rank() ranks.
How is the secret word chosen from an id?	data/repositories/puzzle_repository.dart - puzzleById(); the hash is core/utils/seeded.dart.
What happens on each guess?	features/game/game_controller.dart - submitGuess().
How is typed text cleaned?	core/utils/word_normalizer.dart.
What is saved on the phone?	core/services/storage_service.dart (Hive database), settings_service.dart, data/repositories/stats_repository.dart.
Battle rooms (Firebase side)	features/battle/battle_backend.dart (RtdbBattleBackend talks to Firebase; MemoryBattleBackend is the offline/bot version).
Battle logic and timers	features/battle/battle_controller.dart.
Who is nearest? (scoreboard rule)	features/battle/battle_ranking.dart.
Where to put the Firebase URL	core/constants/backend_config.dart.
Ads	core/services/ad_service.dart, core/widgets/banner_ad_widget.dart, core/constants/ad_ids.dart (test IDs for now).

9. Quick answers

Question	Answer
Does the game need internet?	No - except online Battle rooms.
Is Firebase used right now?	No. It is off until a database URL is added.
Where do answer words come from?	A list built into the app (about 1,850 common nouns).
How do two phones get the same word?	Same id (date or code) through the same formula.
Is the word ever sent online?	Never. Each phone calculates it.
Do two friends see each other's names?	In Battle: yes, live. In Daily: no (not built yet).
Is my progress shared with others?	No. Coins, stats and saved games stay on your phone.

Code: https://github.com/Nikilsuthar/nearword